

Program : Diploma in Instrumentation Engineering	
Course Code : 6082C	Course Title: MEMS and Nano Sensors
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week : 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To gain knowledge of the modern sensor technologies used in industries.
- To acquire knowledge about materials used for manufacturing Microsystems.
- To familiarize manufacturing techniques of MEMS

Course Prerequisites:

Topic	Course code	Course name	Semester
Fundamentals of electronics		Applied physics II	2
Fundamentals of Digital		Fundamentals of electrical and electronics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe the working of micro sensors and actuators.	15	Understanding
CO2	Summarize the materials used for MEMS and Microsystems	15	Understanding
CO3	Explain micro manufacturing techniques	14	Understanding
CO4	Describe the working of nano and bio sensors	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the working of micro sensors and actuators.		
M1.01	Describe micro systems.	2	Understanding
M1.02	Explain about different micro sensors.	5	Understanding
M1.03	Explain the different micro actuation methods.	6	Understanding
M1.04	Describe microfluidics.	2	Understanding

Contents:

Microsystems and microelectronics comparison - Micro sensors features and block diagram-Advantages of Micro sensors – Acoustic wave sensors – Chemical sensors – optical sensors – Pressure sensors – Thermal sensors- Block diagram of micro systems- Micro actuation – Actuation using thermal forces-Actuation using shape memory alloys- Actuation using piezoelectric crystals- MEMS with micro actuators -microgrippers-micromotors-microvalves-micropumps-microaccelerometers – microfluidics.

CO2	Summarize the materials used for MEMS and Microsystems.		
M2.01	Compare substrates and wafers.	1	Understanding
M2.02	Illustrate the crystal structure of silicon.	2	Understanding
M2.03	Summarize various silicon compounds.	6	Understanding
M2.04	Explain different materials used for MEMS and microsystems.	6	Understanding
	Series Test - I	1	

Contents:

Materials for MEMS and Microsystems - substrates and wafers - silicon as a substrate material (the ideal substrate for mems) - crystal structure silicon - silicon compounds - Silicon Dioxide - Silicon Carbide - Silicon Nitride - Polycrystalline Silicon - silicon piezoresistors - gallium arsenide - quartz- piezoelectric crystals - polymers - Conductive Polymers - The Langmuir - Blodgett (LB) film - packaging materials.

CO3	Explain micro manufacturing techniques.		
M3.01	Explain the different etching process	5	Understanding
M3.02	Explain the different etch stop techniques.	2	Understanding
M3.03	Explain the surface micromachining process	3	Understanding
M3.04	Describe the LIGA process	2	Understanding
M3.05	Explain the applications of micro systems.	2	Understanding
Contents: Bulk Micro manufacturing-Etching-Isotropic and anisotropic etching-wet etchants-etch stop-dry etching-Dopant controlled etch stop-Electrochemical etch stop-Dry etching-Plasma etching-Deep reactive ion etching-Surface micromachining-The LIGA process-Major fabrication steps in LIGA process-Applications of microsystems in automotive industry, healthcare industry, aerospace industry, industrial products and consumer products.			
CO4	Describe the working of nano and bio sensors.		
M4.01	Describe the operation of Nano sensors	1	Understanding
M4.02	Illustrate different methods for the measurement of Nanostructure	5	Understanding
M4.03	Explain Nano lithography	4	Understanding
M4.04	Illustrate the working of Carbon Nanotube based sensors	2	Understanding
M4.05	Illustrate the working of Biosensor	2	Understanding
	Series Test - II	1	
Contents: Nano sensors - Measurement of Nanostructure - Probe type sensing-spectroscopic technique - Electrochemical probe - Electron Microscopy - Nano scale Lithography - Carbon Nano Tubes (Basics only) - Carbon Nanotube based sensors - Biosensors - Glucose electrode - electrochemical electrode - Nano scale biosensors for DNA sensing.			

Text / Reference:

T/R	Book Title/Author
R1	Tai-Ran Hsu, MEMS & Microsystems Design and Manufacture, Tata McGraw-Hill Education.
R2	K. Sawhney, A course in Electrical and Electronic Measurements and Instrumentation, Dhanpat Rai and Co, 20th edition.
R3	D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd.
R4	D Patranabis, “ Sensors and Transducers”, PHI.
R5	Ernest O. Doebelin, “Doebelin's Measurement Systems”, McGraw-Hill.

Online Resources:

Sl. No	Website Link
1	https://nptel.ac.in > content > storage2 > courses
2	https://www.sciencedirect.com/topics/engineering/smart-sensors https://www.electronicshub.org > ic-741-op-amp-basics
3	https://nptel.ac.in/content/storage2/courses/downloads/108108125/noc19_ee32_Assignment1.pdf
4	https://www.electronicshub.org/mems-sensors/